

Real-time Domain Adaptation in Semantic Segmentation

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Abstract

State-of-the-art semantic segmentation models can achieve exceptional results on several benchmarks, but they are often computationally complex, leading to high latency and large memory footprint that make them unsuitable for real-world applications. Real-time models aim at optimizing resource demands while sacrificing performance. Furthermore, segmentation datasets are expensive to collect, as they require to annotate large amounts of pixels with the corresponding class labels. Unsupervised Domain Adaptation (UDA) addresses this issue by training models on synthetic datasets, much cheaper to compile, and using a small number of realistic images to allow the models to learn real-world features without using any manually annotated label. In this work we address both problems simultaneously, working on the LoveDA dataset. We apply UDA on PID-Net, a real-time segmentation network, reaching 40 FPS on a NVIDIA Tesla T4 GPU and a mean IoU close to 23%. Additionally, we benchmark other real-time and non-real-time architectures, showcasing the advantages and the downsides of each methods. Finally, we illustrate how the choice of the semantic segmentation loss and the data augmentation techniques enhance or degrades the model performance and its domain generalization capability. The code is available at: <https://github.com/MicheleCazzola/real-time-da-semseg>.

1. Introduction

Semantic segmentation is a computer vision task whose goal is to categorize each pixel of an image into a class. While classification tasks were first solved by larger the receptive fields obtained with deep networks [14], segmentation tasks need spatial information to achieve high-quality results. Tackling this crucial challenge led to several advancements over the last 10 years, resulting in larger and larger models, either performing dense predictions [1, 29] or query-based ones [4, 21]. The most recent architectures

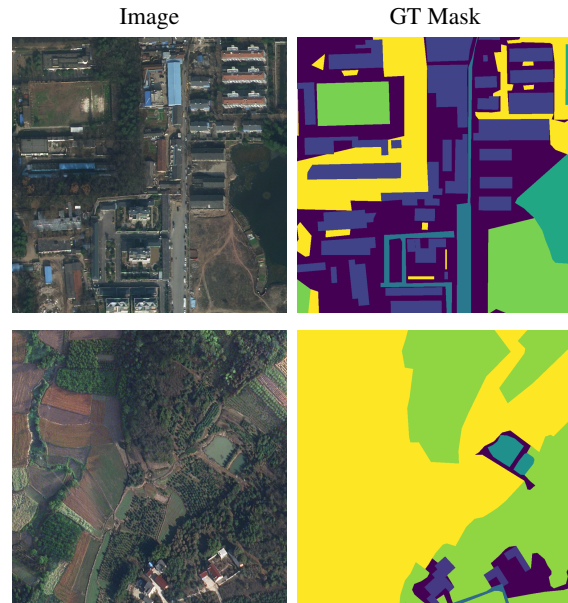


Figure 1. **Qualitative samples of the LoveDA dataset.** The picture in the first row belongs to the urban domain, while that in the second row to the rural one. The main difference between the two domains is the dimension and number of different objects, generally larger for urban samples.

achieve state-of-the-art performance on several benchmarks simultaneously. The immediate downside is related to the latency and memory footprint of such models, which are too large and consequently not suitable for real-time applications such as autonomous driving and industrial manufacturing. For this reason, a parallel research direction consists of developing much smaller segmentation architectures, accepting a lower quality [10, 39]. Despite not reaching state-of-the-art results on various benchmarks, these models satisfy the memory and speed constraint of the aforementioned real-time applications.

Gathering labeled data for semantic segmentation is a time-consuming and costly task, as it requires to employ

several resources to annotate each pixel of the images composing a dataset [7, 25]. It is indeed much easier to synthetically generate such datasets using automated pipelines. This enables to collect larger amounts of data in less time, saving several resources [28, 30].

However, synthetic data come with the downside of being completely different from realistic scenes, representing an obstacle for real-world deployment. For this reason, various domain adaptation strategies have been developed, such as unsupervised domain adaptation (UDA). This task consists of training segmentation models on supervised synthetic data, which is cheap to collect even in large quantity, and using a lower amount of realistic images in an unsupervised setting [27, 34, 35].

In this work, we address both domain adaptation and real-time semantic segmentation. We operate on the LoveDA dataset [38], which explicitly partitions the images in two different domains (urban and rural). Therefore, our goal is to apply PIDNet, a real-time semantic segmentation model, while performing an adversarial-based UDA task [35] between the two domains of LoveDA. This introduces a novelty in the literature as the main domain adaptation approaches often operate on large, high-latency architectures that can provide high-quality segmentation masks.

2. Related work

Semantic Segmentation. Convolutional Neural Networks (CNNs) have successfully solved the problem of image classification, using convolutional filters to extract semantic features and pooling layers to aggregate pixel information [22, 32, 33]. Semantic segmentation requires instead to extract pixel-wise information. Thus, it is necessary to preserve the spatial resolution while retrieving semantic context from the images that plays a crucial role in visual recognition. While early works focused on an autoencoder-like architectures [1, 29, 45] or multi-scale feature aggregation [2, 3, 37, 43], more recent advancements were made leveraging transformer-based architecture [4, 5, 21]. While the former mainly work on extracting the context information while preserving local detailed features, the latter tackle semantic segmentation with a query-based method, using refined multi-scale features to compute the final predictions.

Real-time Semantic Segmentation. Despite the high segmentation performance, the aforementioned techniques often require high computational budget, as a consequence of the elevate number of parameter necessary to extract high-quality context and spatial information. For this reason, several approaches prefer to trade accuracy for speed, being suitable for real-time applications. As in the non-real-time case, they divide in two main families. The first focuses on dense predictions and is often based on

a multi-branch architecture [10, 18, 39–41], while the second focus on designing lightweight transformer-based models [23, 36, 42].

Domain Adaptation. Domain Adaptation methods address the domain shift problem between the source and the target domains by aligning their feature distribution. Early works focused on adversarial learning [11, 12], and were based on GAN-like architectures [15, 44]. Other techniques focused on simpler, fully convolutional approaches, and employ a discriminator to align the domain-specific features in the segmentation maps [16, 26, 35]. More recent works mainly fall under the umbrella of self-training methods, where pseudo-labels for the target domain are generated during training and leveraged to mitigate the domain shift through different adaptation strategies [19, 20, 27, 34].

3. Dataset

Land-cOVER Domain Adaptive Semantic Segmentation (LoveDA) is a dataset for semantic segmentation and domain adaptation tasks. It is composed of 5987 satellite images with 166768 labeled objects belonging to three different Chinese cities. The images have pixel-level semantic annotations and 7 different classes, including road, building, forest, water, agriculture, and barren. An additional background category is introduced to indicate pixels that do not belong to any of the aforementioned classes. Since images can be incomplete or cut, the missing pixels are ignored by default. The dataset is divided into training, validation, and test splits, containing 2522, 1669, and 1796 samples, respectively. Each image belongs either to the urban or the rural domain, that contains 2653 and 3334 elements, respectively [38].

This dataset poses some significant challenges: it contains multi-scale objects, demanding the models to understand features at different resolutions; complex background samples, that represent the majority of the dataset (see Fig. 2) and are not mapped to any real category; inconsistent class distribution across the domains, as shown in Fig. 2. While background cases are widespread in both urban and rural images, the former exhibits only a slight majority of building annotations, and the latter is dominated by agricultural and forest-related samples. Ignored pixels are limited below 5% in both the domains.

The distribution shift between the urban and the rural domain makes this dataset a difficult benchmark for domain adaptation tasks, where a model should infer the pixel-wise labels of samples in a domain despite being trained only on the other.

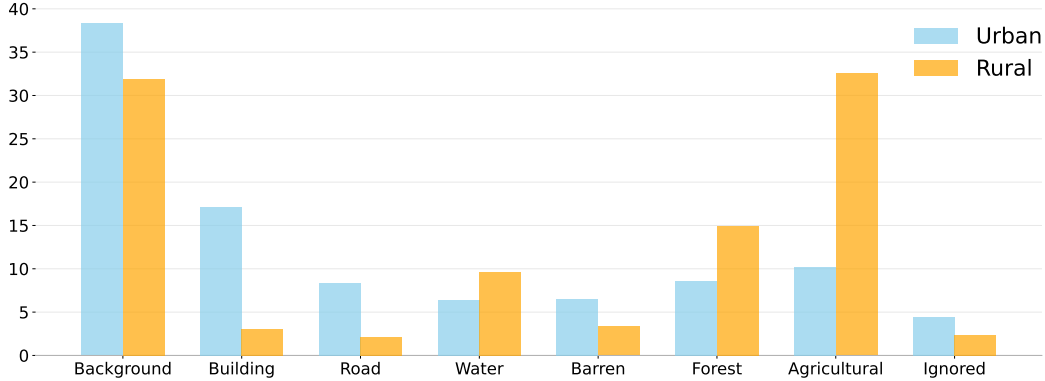


Figure 2. Pixel class distribution in the LoveDA dataset across the urban (light blue) and the rural (orange) domain. The quantities are referred as percents within the corresponding domain. The ignored pixels are reported for completeness.

4. Method

In this work we focus on the Urban $\rightarrow$ Rural adaptation within the LoveDA dataset. Our contribution can be divided in the following three tasks: first, we train real-time and non-real-time segmentation models on the urban domain, and evaluate the performance drop on the rural one; second, we apply adversarial UDA [35] on PIDNet, proving that it mitigates the domain shift gap; third, we study the impact of data augmentation techniques and loss functions on the model performance, highlighting their crucial role.

4.1. Architectures

DeepLabV2. DeepLabV2 is a deep CNN that employs a ResNet101 backbone, followed by an Atrous Spatial Pyramid Pooling (ASPP) module. In the last two layers of the feature extractor, dilated (atrous) convolutions replace max pooling layers, to perform downsampling while better preserving spatial information [17]. The ASPP module is also based on dilated convolutions, but it allows the model to deal with multi-scale reasoning. It utilizes multiple parallel atrous convolutional layers with different dilation rates to capture different spatial and context information [13]. This architectural design allowed DeepLabV2 to achieve state-of-the-art performance on several benchmarks [8, 9].

BiSeNet. Bilateral Segmentation Network (BiSeNet) employs a novel two-branch architecture, composed of a spatial path and a context path. The former is made of three layers, each having a strided convolution followed by batch normalization and ReLU activation. Its role is to extract the output feature maps that at $1/8$ of the original image resolution, preserving the spatial information. The context path is designed to provide sufficient receptive field and thus producing rich semantic features. It employs a lightweight CNN backbone [6, 14], followed by Attention Refinement Modules (ARM) to guide the feature learning

in a multi-scale setting. Finally, the Feature Fusion Module (FFM) concatenates and refines the spatial and context features [41].

STDC. Short-Term Dense Concatenate (STDC) network brings novel enhancements to BiSeNet by removing the structural redundancy and overcoming its two-stream design. The context path is realized in a similar way, using 5 downsampling stages to perform feature extraction. As in BiSeNet, the features from the last two stages are improved using ARMs. Finally, they are fused with the features extracted from the third stage of the backbone using FFM. While BiSeNet learns only from segmentation heads, the third backbone stage is subject to detail guidance. It is realized with an additional detail head trained to learn the boundaries of the objects in the images. Thanks to this refinement, both the context path and the spatial features are aware of detailed boundary information, and this allows STDC to outperform BiSeNet in real-time semantic segmentation with a negligible variation in model size [10].

PIDNet. Proportional-Integrative-Derivative Network (PIDNet) is based on a lightweight, three-branch architecture that takes inspiration from the PID controller and represents an significant novelty with respect to the two-branch networks [10, 18, 41]. An auxiliary derivative branch is added in parallel to the spatial and context path to improve the final segmentation quality. According to an analogy with PID controllers, the three branches have complementary responsibilities: the proportional parses and preserves detailed information in high-resolution feature maps, acting as an all-pass filter; the integral aggregates context information both locally and globally to parse long-range dependencies, acting as a low-pass filter; the novel derivative branch extracts high-frequency features to predict boundary regions, acting as a high-pass

filter [39]. A Pixel-attention-guided (Pag) fusion module is used by the proportional branch to selectively learn the useful semantic features from the integrative branch without being overwhelmed. The features extracted by the context (integrative) branch are refined using the PAPP module, which is a parallel, faster version of DAPP [18] and PPM [43] to fuse multi-scale feature feature maps. The Boundary-attention-guided (Bag) fusion module is instead employed to guide the fusion of detailed (proportional) and context (integrative) representations, given the boundary features extracted by derivative branch.

4.2. Domain adaptation

AdaptSeg. AdaptSeg is an algorithm for adversarial domain adaptation that operates in the output space. It employs two different networks a generator, here represented by the segmentation model, and a CNN-based discriminator, which are trained using a labeled source dataset and an unlabeled target dataset. During training, the generator is optimized using source images and the discriminator prediction from the generator output of the target image. Predictions on source and target image are then given as input to the discriminator. The adversarial approach consists in training the generator to fool the discriminator by generating source outputs aligned with target outputs, and in training the discriminator to fool the generator by correctly discriminating source domain from target domain predictions. The objective function is the weighted sum of the segmentation loss computed on the generator output and the adversarial loss computed on the discriminator predictions [35].

4.3. Loss functions

Cross Entropy loss. Cross-Entropy (CE) loss is a pixel-level loss that approaches 0 as the predicted probability for the target class approaches 1. Given a pixel, CE loss is defined as:

$$L_{CE} = - \sum_{c=1}^C y_c \cdot \log(\hat{y}_c), \quad (1)$$

where C indicates the number of classes, y_c is a boolean flag set to 1 if the pixel belongs to class c , and $\hat{y}_c$ represents the model prediction for the pixel. The pixel-level loss values are averaged on the image samples.

Online Hard Example Mining. Online Hard Example Mining (OHEM) is an algorithm that computes the loss function only on the pixels the model is less confident about. Given a pixel prediction $p \in [0, 1]$, it is selected only if $p < t$, where $t \in (0, 1)$ is an arbitrarily set threshold. To guarantee that the loss statistics are stable across batches, a minimum number of pixels is always employed in the calculation, regardless of the corresponding model confi-

dence [31]. In this work, we applied the OHEM algorithm to the CE loss following [39].

Focal loss. Focal loss dynamically assigns a weight to the CE values. Such weight, ranging in $(0, 1)$, is smaller as the model is more confident on the prediction, and vice-versa. Given a specific pixel, focal loss is defined as:

$$L_{\text{focal}} = - \sum_{c=1}^C y_c \cdot (1 - \hat{y}_c)^\gamma \log(\hat{y}_c), \quad (2)$$

where C , y_c , and $\hat{y}_c$ represents the same quantities as in Eq. (1), while $\gamma > 0$ is a hyperparameter that regulates the entity of the weight. As γ increases the relative loss for well-classified samples decreases. Consequently, the learning process focuses more on correcting misclassifications rather than refining correct predictions [24]. This is similar to the OHEM algorithm, but focal loss employs a smooth weight factor instead of a boolean flag that excludes best-classified samples from the loss computation. The special case $\gamma = 0$ coincides with the CE loss in Eq. (1). As for the CE loss, the pixel-level values are then averaged on the image samples.

5. Experiments

In this section we present the experiments conducted. Specifically, we compare the performance of all the presented models on both the urban and the rural domain, to assess their generalization capability. The effects of domain adaptation are studied on PIDNet only. Finally, the ablation experiments are performed on the three real-time networks and are focused on the effects of augmentation to assess in-domain and cross-domain generalization.

For a fair comparison, we train the models on the training set only and evaluate on the validation set, unless explicitly specified. Following [38], we perform unsupervised domain adaptation using the training split of the source domain and the validation split of the target domain.

5.1. Evaluation metrics

We evaluate the performance of the models based on the mean intersection over union, which is defined as:

$$\text{mIoU} = \frac{1}{C} \sum_{c=1}^C \text{IoU}_c, \quad (3)$$

where C represents the number of classes and IoU the within-class intersection over union. Such metric is defined as:

$$\text{IoU}_c = \frac{\sum_{i=1}^N |\{\hat{y}_{i,p} = y_{i,p}\}_{p=1}^{P_i} \cap \{y_{i,p} = c\}_{p=1}^{P_i}|}{\sum_{i=1}^N |\{\hat{y}_{i,p} = y_{i,p}\}_{p=1}^{P_i} \cup \{y_{i,p} = c\}_{p=1}^{P_i}|} \quad (4)$$

Model	Backbone	Urban	Rural	Domain gap	FPS	GFLOPs	Params
DeepLabV2	ResNet101	29.5	21.1	8.4	2.0	728.9	43.0M
BiSeNet	ResNet101	<u>31.7</u>	21.1	10.5	6.2	177.6	48.3M
BiSeNet	ResNet18	30.1	18.7	11.4	24.0	49.4	11.9M
STDC1-Seg	-	33.5	22.2	11.3	30.3	33.8	12.0M
PIDNet-S	-	31.1	<u>21.2</u>	10.0	39.5	25.3	7.7M

Table 1. **Model comparison on mean IoU and resource consumption.** The central section represents the urban and rural mIoUs (%) and the corresponding gap associated with domain shift. The right section shows the speed and resource-related metrics, computed on a Tesla T4 GPU with resolution 1024×1024 and batch size 1. The best results are highlighted in **bold**, the second best are underlined.

where i and p index an image and a specific pixel respectively, while N and P_i indicate the total number of images and the total number of pixels for the i -th image. $\hat{y}_{i,p}$ and $y_{i,p}$ represent the prediction and ground truth for the p -th pixel of the i -th image.

Furthermore, we underline the importance of resource- and time-consumption by measuring the number of parameters, the floating point operations per second (FLOPs), and the frames per second (FPS) for each model.

5.2. Implementation details

We conduct our experiments on a NVIDIA Tesla T4 GPU in the Google Colab environment, with 15 GB of available memory. We measure the FLOPs and the FPS of the models at resolution 1024×1024 with batch size 1.

Due to the limited computational capability, we limit the number of training epochs to 20 for the model comparison task, to allow the larger networks to be trained in reasonable time. For the same reason, we trained the models with batch size 6. Furthermore, we apply random crop augmentation with size 512×512 . The models are pretrained on ImageNet, following [2, 10, 39, 41]. We use AdamW optimizer with learning rate 10^{-3} and 10^{-4} for the fresh and pretrained parameters, respectively. Weight decay is set to 10^{-3} . We adopt a polynomial scheduler with exponent 0.9. We use the OHEM loss to allow the model to learn difficult samples with higher priority.

For the adversarial domain adaptation task, we start training from the ImageNet pretrained PIDNet-S. Following [38], we use cross-entropy loss and AdamW with batch size 8, with a base learning rate of $1.25 \cdot 10^{-4}$, which is multiplied by 10 for fresh parameters only. The weight decay and scheduling are the same as the within-domain training. We employ the fully convolutional discriminator described in [35], optimized with Adam featuring a learning rate of 10^{-4} and beta coefficients (0.9, 0.99). The learning rate is decayed using the same scheduler of the generator network. Contrarily to previous works, we find the best result with adversarial coefficient λ of 10^{-5} . Finally, we employed both random crop augmentation for each sample and one of random mirroring and orthogonal random rotation with proba-

bility 0.75. For fair comparison, we train PIDNet-S on the urban domain using the same schedule.

In the ablation studies, we show how the choice of the loss function and of the augmentation influences the results, focusing on the PIDNet-S model.

5.3. Experimental results

Within-domain semantic segmentation. We compared the aforementioned models on the semantic segmentation task, focusing on the urban domain of LoveDA. As shown in Tab. 1, STDC-Seg presents the higher mIoU (33.5%), followed by the ResNet-101 BiSeNet and PIDNet-S. As expected, the real-time BiSeNet based on ResNet18 pays 1.6 points to its larger version, while DeepLabV2 does not even reach 30%. Furthermore, DeepLabV2 is the slowest analyzed model, while PIDNet the fastest, reaching 20x the FPS of DeepLabV2. Moreover, STDC-Seg and BiSeNet represent valid real-time alternatives, both reaching more than 20 FPS. We observe the same trend in both the FLOPs and the number of parameters. The only exception is that ResNet101-BiSeNet is larger than DeepLabV2, despite being three times faster.

Zero-shot domain shift. We evaluate the models on the rural domain, to assess the cross-domain generalization without any domain adaptation from the urban domain. As presented in Tab. 1, STDC1-Seg achieves the best performance, with a 22.2% mIoU, resulting in a 11.3 points of domain gap. Similar degradation is observed for the real-time BiSeNet. PIDNet-S, ResNet101-BiSeNet and DeepLabV2 obtain a similar mIoU around 21%, with the latest presenting the smallest domain gap of 8.4%. Despite having lower performance with respect to STDC1-Seg, PIDNet showcases nearly equivalent mIoU of larger models, featuring much higher speed (from 6 to 20 times) and a smaller memory footprint (around 6 times fewer parameters). We are aware that a much reliable comparison should involve larger batch size and more training epochs, which can be affordable only with unlimited and higher-range hardware. However, these results clearly showcase the speed and memory advantage of PIDNet-S with respect to single-branch

Model	Background	Building	Road	Water	Barren	Forest	Agriculture	Average
Urban → Urban								
DeepLabV2	31.0	41.5	38.3	36.0	8.8	31.7	19.4	29.5
BiSeNet-R101	33.3	45.6	42.6	38.1	8.6	32.3	21.2	<u>31.7</u>
BiSeNet-R18	31.7	<u>42.3</u>	37.6	<u>38.2</u>	9.2	<u>33.3</u>	18.4	30.1
STDC1-Seg	<u>32.7</u>	42.0	41.1	40.7	<u>11.8</u>	35.8	30.6	33.5
PIDNet-S	30.3	39.8	<u>41.6</u>	34.5	12.7	32.8	<u>26.1</u>	31.1
Urban → Rural								
DeepLabV2	43.5	19.7	12.1	<u>30.8</u>	5.8	9.2	26.6	21.1
BiSeNet-R101	45.8	25.0	<u>13.6</u>	25.2	2.0	7.2	<u>29.1</u>	21.1
BiSeNet-R18	42.4	24.5	10.3	21.0	2.7	8.4	21.4	18.7
STDC1-Seg	<u>43.9</u>	26.2	12.3	32.0	3.6	7.2	30.2	22.2
PIDNet-S	43.6	<u>25.4</u>	14.4	23.5	<u>5.6</u>	<u>8.9</u>	26.8	<u>21.2</u>

Table 2. **Model comparison on per-class and mean IoU.** Urban and rural results are computed after training the models on the urban domain. The results are reported as a percentage (%). For each class and target domain, the best result is highlighted in **bold**, while the second best is underlined.

Model	Background	Building	Road	Water	Barren	Forest	Agriculture	Average
Oracle	32.3	42.5	42.9	38.6	11.7	35.8	27.8	33.1
Source only	45.8	28.3	13.8	22.9	6.2	6.1	27.3	21.5
AdaptSeg [35]	46.4	27.0	14.1	33.4	8.1	5.0	25.2	22.7

Table 3. **Adaptation strategies comparison on per-class and mean IoU with PIDNet-S.** The first and second row represent the PIDNet-S model trained on the urban domain and evaluated on the urban and rural one, respectively.

and double-branch architectures, without significant performance degradation.

Per-class performance. We study the detailed performance of the models on each class, in both the experimental settings. Tab. 2 shows that there does not exist an architecture that clearly outperforms the others. In the within-domain task, STDC showcases more robustness across classes, achieving top-2 performance in 5 out of 7 cases, with limited degradation in the other cases. BiSeNet-R101 showcases the highest result on background (33.3%), building (45.6%), and road (42.6%) elements. STDC works best on water (40.7%), forest (35.8%) and, most clearly, agriculture (30.6%) classes. Furthermore, it does not lose significant accuracy to BiSeNet-R101 in the other cases, despite using much fewer parameters. PIDNet-S, the faster and smaller architecture, achieves the highest IoU on barren regions (12.7%), but suffers some degradation on the other classes, such as building (39.8%), water (34.5%), and agriculture (26.1%).

The detail on the rural domain shows a trend common to all the models. While background and the agriculture classes are better recognized, the others are subject to relevant degradation. As for the urban domain, STDC is the

most robust model, showcasing the best performance on building (26.2%), water (32.0%), and agriculture (30.2%) regions, and the second best on background cases (43.9%). PIDNet achieves the highest IoU on road pixels (14.4%), while not suffering of significant degradation on background, building, barren and forest regions with respect to the best performing model. However, PIDNet pays 8.5 and 3.4 points to STDC on water and agriculture classes, similarly to what observed for the urban domain. Generally, we hypothesize that the background and agricultural samples are easier to recognize in the rural domain, as they may be not interleaved by complex and small anthropic structures such as buildings and roads. Conversely, we think that the other objects present themselves with significantly different characteristic in the two domains, leading to the observed performance drop.

Adversarial domain adaptation. We applied adversarial domain adaptation (AdaptSeg) on PIDNet, following [35, 38]. As illustrated in Tab. 3, AdaptSeg improves the average result by 1.2 points, achieving 22.7% mean IoU on the rural domain. Furthermore, the adversarial strategy allows PIDNet to improve the IoU on background (+0.6), road (+0.3), barren (+1.9) and most significantly water (+10.5) regions.

Model	Aug.	Urban	Rural	Domain gap
BiSeNet-R18	✓	26.7	18.2	8.5
		30.1	18.7	11.4
STDC1-Seg	✓	27.7	19.2	8.5
		33.5	22.2	11.3
PIDNet-S	✓	27.6	18.8	8.8
		31.1	21.2	9.9

Table 4. **Ablation on the random crop augmentation.** The lines without a check mark indicate experiments where the training images were directly resized to 512×512 resolution. Models were trained with OHEM loss.

However, this method presents evident limitations, causing a degradation on building (-1.3), forest (-1.1) and agricultural (-2.1) samples.

5.4. Ablation studies

We conduct some ablation studies to highlight the impact of the loss and of the augmentation techniques on the segmentation result. We assess the model quality both on the urban and the rural domain, after training for 20 epochs on the urban images, using the same hyperparameter and schedule described in Sec. 5.2.

Random crop augmentation. We investigate how randomly cropping training samples influences the performance on the source and on the target domain. As shown in Tab. 4, random crop augmentation is beneficial for semantic segmentation on both domains across the three tested models. While urban mIoU is affected by an increase from 3.5 to 5.8%, the rural mIoU improvement is more modest. In particular, it ranges from a minimum of 0.5% for the ResNet18-BiSeNet to 30% on the STDC1-Seg model. The domain gap assumes similar values across the architectures. This outcome suggests that a simple random crop augmentation enhances generalization both in the source and in the target domain.

Loss functions. We study the impact of the loss function on the model generalization capability, and how it relates with the random crop augmentation. Tab. 5 illustrates the results with cross-entropy (CE), OHEM, and focal loss on the PIDNet-S model. While cross-entropy performs best without any augmentation (27.8% urban mIoU), OHEM represents the best loss function under random crop augmentation (31.1 % mIoU). In both cases, the domain shift gap ranges between 8 and 10%. Focal loss represents the worst performing case on both domains regardless the augmentation being used, with a mIoU degradation between 0.5 and 1.5% with respect to the other methods.

Loss	Aug.	Urban	Rural	Domain gap
CE	✓	27.8	18.6	9.2
		31.0	22.6	8.4
OHEM	✓	27.6	18.8	8.8
		31.1	21.2	9.9
Focal	✓	26.7	17.3	9.4
		30.2	20.6	9.6

Table 5. **Ablation on the loss function.** The lines without a check mark indicate experiments where the training images were directly resized to 512×512 resolution. Results refer to the PIDNet-S model.

Loss	RC*	C [†]	Urban	Rural	Domain gap
CE	✓	✓	31.0	22.6	8.4
	✓		30.7	21.2	9.5
OHEM	✓	✓	31.1	21.2	9.9
	✓		30.3	19.6	10.7
Focal	✓	✓	30.2	20.6	9.6
	✓		30.0	19.2	10.8

Table 6. **Ablation on the compound augmentation.**[†]: random crop.[†]: compound augmentation. Results refer to the PIDNet-S model.

Other geometric augmentations. We assess the effectiveness of the compound augmentation widely used in this task [35, 38, 39], which consists of applying one among random mirroring and random rotation after the random crop. Since we are working on satellite images, the random rotation is set to an angle multiple of 90° . Tab. 6 shows the effect of this augmentation, which are generally unfavorable across the loss functions used. Although the degradation is limited on the urban domain, the rural one shows a drop between 1 and 2%. In addition, the domain shift gap undergoes an increase ranging from 0.5 to 1.1%. As this further augmentation technique causes such performance degradation, we may conclude that it introduces much more data variability than what PIDNet-S can handle. This may result in an over-regularized model, that not only perform worse on validation samples, but also overfits on training data. However, we are aware that more accurate comparison may be performed with a much longer training schedule and a larger batch size.

6. Conclusions

In this work, we analyzed and investigated domain adaptation techniques for real-time semantic segmentation. Specifically, we focused on PIDNet, a low-latency

model that achieves competitive results on the LoveDA dataset. Furthermore, we addressed the domain shift problem through adversarial domain adaptation, which can effectively mitigate the degradation with a negligible amount of additional memory. Finally, we studied the effects of loss functions and data augmentation techniques on the model performance, highlighting their crucial role in semantic segmentation and domain generalization.

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